

Hyunseok Seung
1207 University Avenue
Madison, WI 53706
Email: hseung2@wisc.edu | Website: hseung88.github.io

Professional Appointments

- 2025 – Present **Postdoctoral Research Associate**, *Department of Statistics, University of Wisconsin–Madison*, Madison, WI
Advisor: [Matthias Katzfuss](#)
- 2018 **Data Scientist**, *SK hynix Inc.*, Icheon, South Korea

Education

- 2025 **Ph.D.** in Statistics, *University of Georgia*, Athens, GA
Advisors: [Jaewoo Lee](#) and [Yuan Ke](#)
Dissertation: *Scalable and Efficient Learning: Algorithmic Advances for Time Series and Deep Neural Models*
- 2018 **M.A.** in Applied Statistics, *Yonsei University*, Seoul, South Korea
Advisor: [Sangun Park](#)
Thesis: *Modified Likelihood Ratio Tests for Extreme Value Distributions*
- 2016 **B.A.** in Applied Statistics, *Yonsei University*, Seoul, South Korea

Research Interests

Scalable probabilistic machine learning with gradients and curvature:

- Scalable Gaussian processes with gradient observations
- Bayesian optimization and sample-efficient decision making
- Memory-efficient fine-tuning of large language models
- Curvature-aware optimization for deep neural networks

Publications

Conference Proceedings

- C1. **Hyunseok Seung** & Katzfuss, M. *Scalable Derivative Gaussian Processes via Exact Gradient Reduction* in *Advances in Neural Information Processing Systems (NeurIPS)* [[ArXiv](#)] (2026).
- C2. **Hyunseok Seung**, Lee, J. & Ko, H. *Low-Rank Curvature for Zeroth-Order Optimization in LLM Fine-Tuning* in *AAAI Conference on Artificial Intelligence* [[DOI](#)] (2026).
- C3. **Hyunseok Seung**, Lee, J. & Ko, H. *MAC: An Efficient Gradient Preconditioning using Mean Activation Approximated Curvature* in *IEEE International Conference on Data Mining (ICDM)* [[DOI](#)] (2025).

- C4. **Hyunseok Seung**, Lee, J. & Ko, H. *NysAct: A Scalable Preconditioned Gradient Descent using Nystrom Approximation* in *IEEE International Conference on Big Data* [DOI] (2024).
- C5. **Hyunseok Seung**, Lee, J. & Ko, H. *An Adaptive Method Stabilizing Activations for Enhanced Generalization* in *IEEE International Conference on Data Mining Workshop (ICDMW)* [DOI] (2024).

Journal Articles

- J1. **Hyunseok Seung**, Han, K., Shen, Y. & Ke, Y. Enhancing COVID-19 Mortality Prediction with Online Autocovariance Change Points Detection. *Stat* **15**. [DOI] (2026).
- J2. **Hyunseok Seung**, Lee, J. & Ko, H. Mean Activation Curvature for Scalable Second-Order Optimization in Deep Networks. *Knowledge and Information Systems*. [DOI] (2026).
- J3. **Hyunseok Seung** & Park, S. Modified Likelihood Ratio Tests for Extreme Value Distributions. *Communications in Statistics–Theory and Methods* **52**. [DOI], 5742–5751 (2021).

Research Experience

Core Research Areas

- 2025 – Present **Postdoctoral Researcher**, *Department of Statistics, University of Wisconsin–Madison*
Scalable Gaussian Processes with Derivatives (advised by [Matthias Katzfuss](#))
- Developed an exact gradient reduction principle for derivative Gaussian processes, enabling scalable function-value prediction from gradient observations while preserving the full gradient information within each Vecchia conditional.
 - Developed an approximation for derivative Gaussian processes using two directional derivatives per observed gradient, closely matching the posterior obtained from full gradients while further reducing computational and memory costs relative to exact gradient reduction.
- 2022 – 2025 **Research Assistant**, *School of Computing, University of Georgia*
Curvature-Aware Optimization for Deep Learning (advised by [Jaewoo Lee](#))
- Developed a curvature-aware zeroth-order optimizer for LLM fine-tuning, improving convergence and test accuracy over MeZO-Adam while reducing memory usage by up to 27%.
 - Developed scalable second-order optimization methods using activation covariance, improving vision transformer test accuracy by 3.6 percentage points over KFAC.

Collaborative and Earlier Research

- 2025 – Present **Postdoctoral Researcher**, *Department of Statistics, University of Wisconsin–Madison*
Hyperspectral Foundation Models (collaboration with [Townsend Lab](#))
- Developing hyperspectral foundation models by pre-training vision transformers on hyperspectral data and fine-tuning them for downstream trait prediction, with conformal prediction used for uncertainty quantification.
- 2023 – 2024 **Research Assistant**, *Department of Educational Psychology, University of Georgia*

Topic Modeling (collaboration with [NPDA Lab](#))

- Analyzed video and text data using automatic speech recognition and topic modeling, collaborating with researchers in mathematics education and psychology.

2021 – 2023

Research Assistant, *Department of Statistics, University of Georgia*

Time Series Forecasting (advised by [Yuan Ke](#))

- Developed hybrid COVID-19 mortality forecasting models using online autocovariance change point detection, improving prediction accuracy by 6% and reducing training time by 99% compared to standard rolling-window cross validation.

2018

Data Scientist, *SK hynix Inc.*

Wafer Failure Early Detection System (supervised by [Sangun Park](#))

- Identified key predictors of wafer failure using statistical models for high-dimensional semiconductor fabrication data.

2017 – 2018

Research Assistant, *Department of Applied Statistics, Yonsei University*

Modified Likelihood Ratio Tests (advised by [Sangun Park](#))

- Developed modified likelihood ratio test statistics tailored to highly skewed settings to improve sensitivity to tail departures.

Talks and Presentations

Oral

- T1. *Scalable Derivative Gaussian Processes via Exact Gradient Reduction* Institute for Foundations of Data Science (IFDS) Research Expo (Madison, WI, USA). Aug. 2026.
- T2. *An Efficient Gradient Preconditioning using Mean Activation Approximated Curvature* IEEE International Conference on Data Mining (ICDM) (Washington, DC, USA). Nov. 2025.
- T3. *A Scalable Preconditioned Gradient Descent using Nystrom Approximation* IEEE International Conference on Big Data (Washington, DC, USA). Dec. 2024.
- T4. *Modified Likelihood Ratio Tests for Extreme Value Distributions* The Korean Data and Information Science Society (KDISS) (Pukyong University, Busan, South Korea). May 2018.

Posters

- P1. *Low-Rank Curvature for Zeroth-Order Optimization in LLM Fine-Tuning* AAAI Conference on Artificial Intelligence (Singapore). Jan. 2026.
- P2. *A Scalable Preconditioned Gradient Descent using Nystrom Approximation* AI Research Day, Institute for Artificial Intelligence (Athens, GA, USA). Apr. 2025.
- P3. *An Adaptive Method Stabilizing Activations for Enhanced Generalization* AI Research Day, Institute for Artificial Intelligence (Athens, GA, USA). Apr. 2024.
- P4. *Modified Likelihood Ratio Tests for Extreme Value Distributions* The Korean Statistical Society (KSS) (Pusan National University, Busan, South Korea). May 2018.

Software

2026	TERA Exact gradient reduction for scalable derivative Gaussian processes Official PyTorch implementation of target-specific directional reduction that preserves the predictive information in gradient observations while avoiding full gradient covariance computations. [Code]
2025	LOREN Low-rank curvature for zeroth-order LLM fine-tuning Official PyTorch implementation of low-rank curvature-aware zeroth-order optimization with RLOO variance reduction for memory-efficient LLM fine-tuning. [Code] MAC Mean-activation curvature for scalable second-order optimization Official PyTorch implementation of rank-one mean-activation curvature preconditioning for scalable training of convolutional neural networks and vision transformers. [Code] CP Change-point-adaptive COVID-19 mortality forecasting Research code combining online autocovariance change-point detection with national- and state-level mortality forecasting pipelines. [Code]
2024	NysAct Nyström activation-covariance preconditioning Official PyTorch implementation of Nyström-approximated activation covariance preconditioning for scalable curvature-aware training of convolutional neural networks and vision transformers. [Code] AdaAct Activation-variance-aware adaptive optimization Official PyTorch implementation of adaptive optimization based on activation variance for stabilizing learned representations and improving generalization. [Code]

Teaching Experience

2018	Lecturer, Instructor of Record, Yonsei University STAT1001 Introduction to Statistics	Fall 2018
2019 – 2023	Teaching Assistant, University of Georgia <i>Graduate Courses</i> STAT6430 Design and Analysis of Experiments STAT6315 Statistical Methods for Researchers STAT8330 Advanced Statistical Applications and Computing STAT6420 Applied Linear Models STAT8440 Statistical Inference in Bioinformatics STAT6210 Introduction to Statistical Methods STAT6210 Statistical Methods <i>Undergraduate Courses</i> STAT4230 Applied Regression Analysis STAT2360 Programming and Data Literacy using R STAT4210 Statistical Methods STAT3110 Introduction to Statistics for Life Sciences STAT3120 Introduction to Probability for Life Sciences	Spring 2023 Spring 2023 Fall 2022 Fall 2022 Fall 2020 Fall 2020 Fall 2019 Spring 2022 Fall 2021 Spring 2021 Summer 2020 Spring 2020
2017 – 2018	Teaching Assistant, Yonsei University STAT1001 Introduction to Statistics STAT1001 Introduction to Statistics STAT1001 Introduction to Statistics	Spring 2018 Fall 2017 Spring 2017

Honors & Awards

2018	Industry-Sponsored Scholarship Recipient, <i>SK hynix Inc.</i>
2018	Best Paper Presentation Award, <i>Korean Data and Information Science Society</i>
2017 – 2018	Brain Korea 21 Fellowship, <i>Korean Government-Funded Graduate Program</i>
2015	Honors Award, <i>Yonsei University</i>

Academic Service

Reviewer

2026	International Conference on Learning Representations (ICLR 2027)
	Journal of the American Statistical Association (JASA), Theory and Methods
	Journal of Computational and Graphical Statistics (JCGS)
	Advances in Neural Information Processing Systems (NeurIPS 2026)

Registered Student Organization Advisor

2026 – 2027	KoJobs UW–Madison, University of Wisconsin–Madison
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Last updated: September 24, 2026